

QSARena: model-library breadth drives most leaderboard standing

One leakage-controlled pipeline, 44 datasets, 28 models - and no model family dominates.

BENCHMARK



SMILES
to property

44 datasets

5 suites

28 models

TDC | MoleculeNet | Polaris
PODUAM | ChemML

ESTIMATED TOP-10 VS PUBLISHED VALUES (PROVISIONAL)

Best of 28 models, chosen on the test set



22/22 on official TDC splits - median rank 3

Same protocol, cross-validation-eligible models only



median rank 6

Chosen by cross-validation only - the honest estimate



median rank 8 - and zero first places

-7
model-library
breadth

-3
held-out
selection

NO SINGLE WINNER

Ensembles 16

3D pretrained 11

Conventional ML 8

Top family takes only 36% of datasets;
the best model is dataset-dependent.

RUNS ON A LAPTOP GPU

-0.2% median change vs an A100,
37 identically split datasets

Start with descriptor-rich gradient boosting plus averaging; add 3D pretrained models where chemistry and compute justify it.

github.com/ScottCoffin/QSARena